

TRANSISTOR TESTER

AIM: To design and test transistor tester using Arduino NANO

Components Required:

SI.NO.	Description	Quantity
1.	Arduino NANO board	1
2.	LCD 16x2 Display	1
3.	470-ohm Resistor	3
4.	Push button	1
5.	Bread Board	1
6.	Jumping wires	9

ARDUINO NANO:

Arduino Nano is one type of microcontroller board, and it is designed by Arduino.cc. It can be built with a microcontroller like Atmega328. It is a small size board and also flexible with a wide variety of applications. Other Arduino boards mainly include Arduino Mega, Arduino Pro Mini, Arduino UNO, Arduino YUN, Arduino Lilypad, Arduino Leonardo, and Arduino Due. And other development boards are AVR Development Board, PIC Development Board, Raspberry Pi, Intel Edison, MSP430 Launchpad, and ESP32 board.

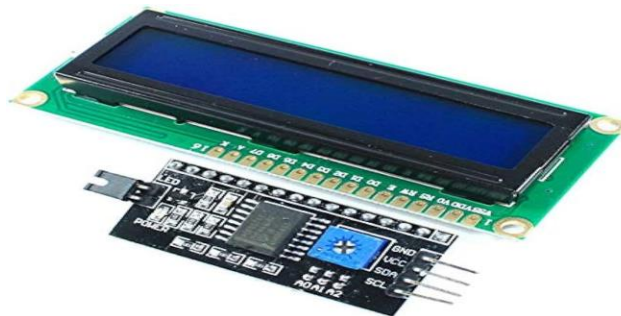
This board has many functions and features like an Arduino Duemilanove board. However, this Nano board is different in packaging. It does not have any DC jack so that the power supply can be given using a small USB port otherwise straightly connected to the pins like VCC & GND. This board can be supplied with 6 to 20volts using a mini-USB port on the board.

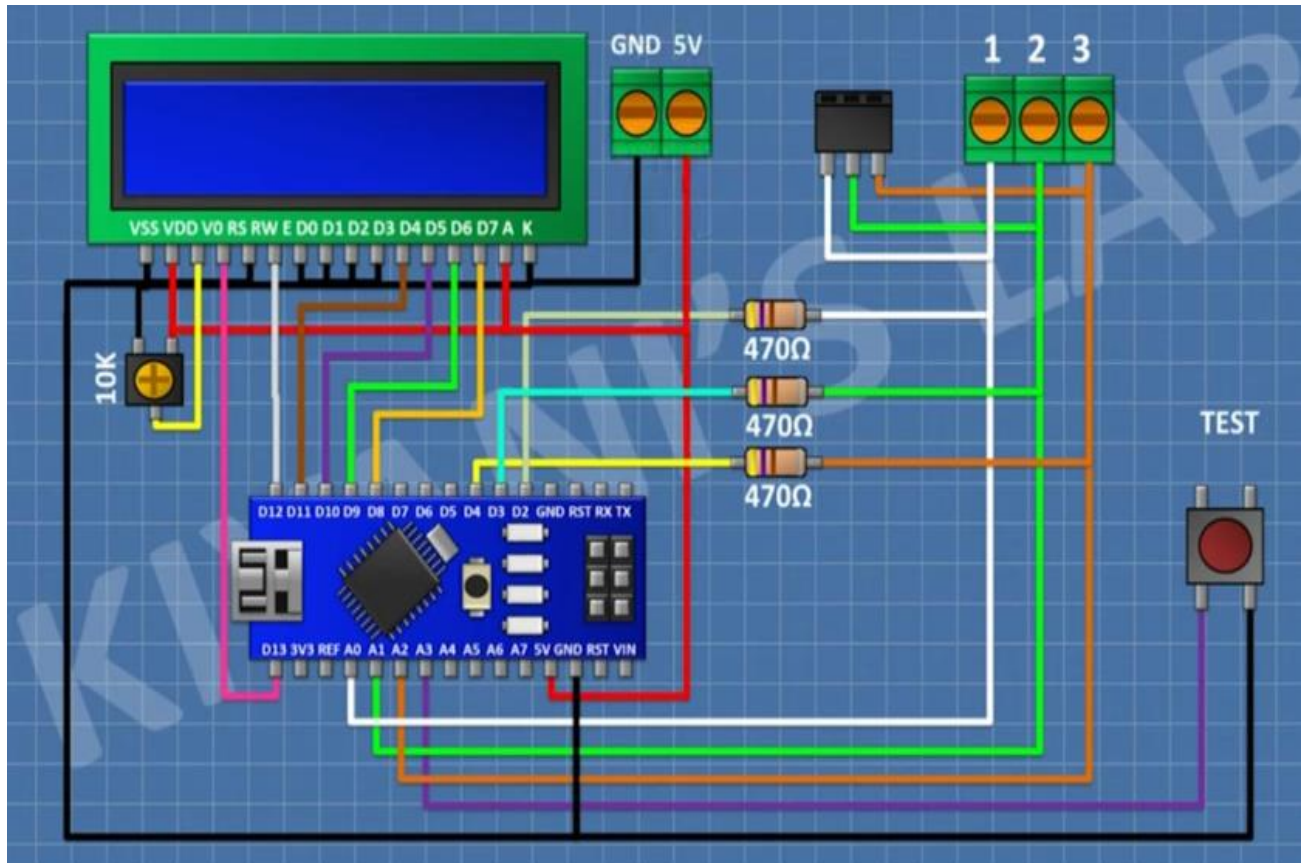


LCD 16X2 DISPLAY:

The term LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations, etc.

An I2C LCD is a Liquid Crystal Display that uses the I2C (Inter-Integrated Circuit) protocol to communicate with microcontrollers like the Arduino. It significantly reduces the number of pins needed to control the display—from 6 or more to just 2 (SDA and SCL).



CIRCUIT DIAGRAM:**WIRING CONNECTION:****LCD I2C Display:**

GND Ground

VCC Power (5V)

SDA Data Line – A4 on NANO

SCL Clock Line – A5 on NANO

Push Button: 1-A3, 2-GND

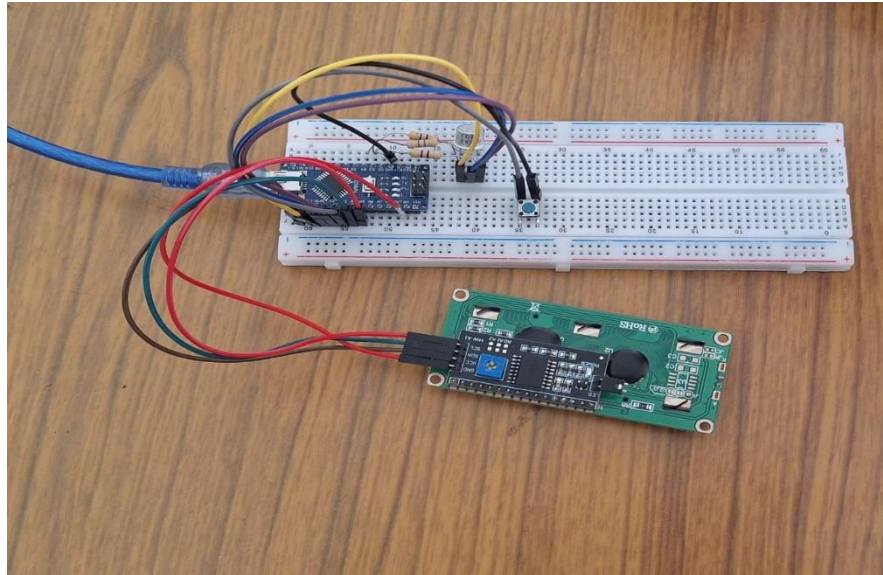
Resistor:

1-D2, 2-D3, 3-D4

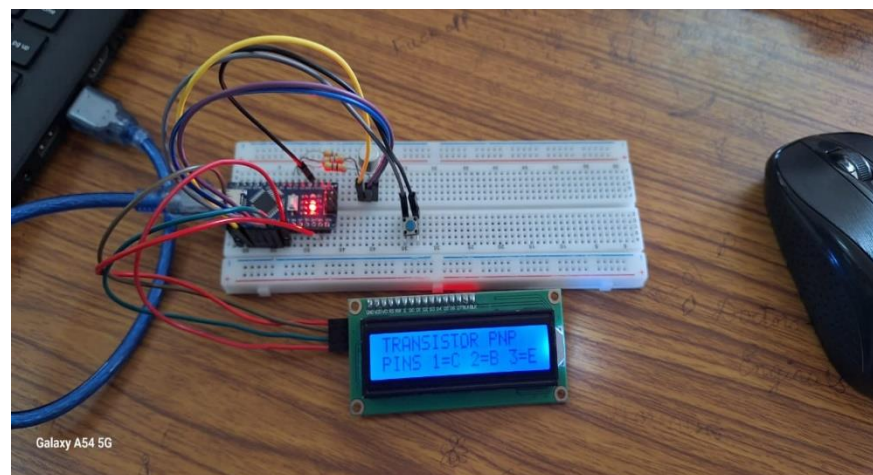
D2 Second leg – A0

D3 Second leg – A1

D4 Second leg – A2



EXPERIMENTAL RESULT



CONCLUSION:

The transistor tester proved to be an essential and practical tool for identifying and analyzing various types of transistors. Through this project we successfully demonstrated the ability of the transistor to determine the type (NPN or PNP), pin configuration of Bipolar Junction Transistor. The tester provides the quick and accurate method for transistor verification, greatly aiding in troubleshooting and circuit development. Its ease of use, low cost, and reliability make it an invaluable asset for both hobbyists and professionals in the field of electronics. Overall, the transistor enhances efficiency and accuracy in electronic diagnostics and design.